

“WHO'S TAKING WHAT?” CONNECTING NEUROSCIENCE, PSYCHOPHARMACOLOGY AND INTERNAL FAMILY SYSTEMS FOR TRAUMA

Frank Guastella Anderson

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INTRODUCTION

This chapter is focused on integrating psychopharmacology for acute or chronic complex trauma with psychic multiplicity and Self-energy, the central themes of internal family systems (IFS) therapy. In the service of this goal I developed five main strategies for working with clients. First I identify a list of symptoms, keeping in mind that *medications treat symptoms not feelings*. Second, I open a dialogue with the client's parts, to help differentiate a part response from a biological condition. This dialogue with parts helps me to be clear about what I am treating and helps me track symptoms so that I am sure the medications are actually working. Third, I validate the experience parts have had with medication and address any future concerns. Here my policy is *no medications are prescribed unless all parts are in agreement*. Obstacles to reaching consensus are common and will be discussed later. Fourth, I educate parts about what to expect from medications and maintain open communication with them throughout the prescribing process. My fifth and last strategy is to be aware of my own parts as they arise while prescribing. I developed a policy of *I educate, you decide*, which helps my parts to step back and make space for Self-energy. In sum, I *identify* symptoms, *differentiate* emotional from biological,

validate concerns, *educate*, *collaborate* and *monitor my parts*. These strategies are illustrated with case examples later but first, because the neurobiology of trauma must guide medication choices, here is a review of the relevant basics of neuroscience and neurobiology.

NEUROSCIENCE

Neurons in the brain strengthen and grow new synapses when activated by experience. *Neuroplasticity* refers to connections changing in response to experience (Doidge, 2007). When these neurons fire, new nerve cells can grow. This is called *neurogenesis* (Doidge, 2007). Experience thus creates structural change in the brain. *Neural integration* is a process of coordination and balance between different brain regions that are functionally linked (Siegel, 2011). For neural integration to occur, the brain needs to function optimally, with all parts working together in harmony so that new neural networks can form and behavioral change can take place. Because trauma affects several key areas of the brain, recovery involves reestablishing balance. The first area affected by trauma is the brainstem, the most primitive part of the brain, which receives input from the body and regulates heart rate, breathing, alertness, sleep and the fight-flight-freeze response (Siegel, 2007). The second area that is affected, the limbic system, is composed of the amygdala, hippocampus and hypothalamus. This area controls affect, attachment, memory and meaning making (Siegel, 2007). The third area affected by trauma is the cortex, which includes the right and left hemispheres and the prefrontal cortex. The right hemisphere deals with perception, holistic, nonverbal and autobiographical memory whereas the left hemisphere is factual, linear, logical and deals with language (Siegel, 2007). The right hemisphere takes information from the body and brainstem and integrates it with the left hemisphere (Schore, 2003). The prefrontal cortex, a key structure involved in mindfulness, is vital in coordinating all the areas of the brain mentioned above to function as a whole (Siegel, 2007). The anterior cingulate within the prefrontal cortex is involved in emotional processing and working memory (Siegel, 2007). *Mirror neurons*, crucial to empathy and connection, also reside in the cortex (Siegel, 2010). The therapeutic protocol of IFS, with its mindful focus on parts, helps protectors relax so that Self-energy can emerge. This can create the optimal environment for the brainstem, limbic system, cortex and especially the pre-frontal cortex to work together, promoting neural integration and creating new pathways in the brain.

TRAUMA AND MEMORY

Memory is either implicit or explicit. Implicit memory occurs within the first 18 months of life and is primarily focused on perception, emotion and bodily sensation (Siegel, 2011). It functions to generalize experience and primes the brain for future action (Siegel, 2011). It does not involve focal, conscious attention and it lacks the awareness that comes from the past (Siegel, 2011). Many memories and reenactments of trauma come from implicit memory and are often expressed by firefighters or exiles.

For example, Jane came into a session after being stuck in a hotel room on a business trip, unable to return home because of a tornado. She was describing the storm when a train went by, a common occurrence at my office, where Jane had been many times. But this time she threw her coat over her head and screamed, “Make it stop, help me, please help me!”

Not sure what was going on, I said, “Jane try to look at me. Can you hear my voice? See if you can open your eyes to see me and see where we are. Look me in the eye. You are safe here.”

Jane looked at me and then around the office for a few minutes and gradually calmed down. “The sound of the train was like the tornado,” she finally explained.

For those moments Jane had been re-experiencing the roaring of the tornado bearing down on her. She was flooded with feelings and sensations identical to that time and was totally unaware of her present surroundings. When a client is blended with a traumatized part in this way, perspective and context disappear and she is engulfed in the past. Jane had no words, no narrative to attach to the experience—implicit memory was taking over her brain.

Explicit memory starts to develop at age 2 years. Requiring focused attention, this form of memory is factual and episodic (Siegel, 2011). Because explicit memory includes the hippocampus, it involves a sense of time and narrative (Siegel, 2011). A major goal of trauma therapy and of the process of unburdening in IFS is to turn implicit into explicit memory, bringing a cohesive narrative to traumatic experience. Some protectors have qualities of implicit memory. Firefighters, often extremely intense, reflexive and all- or-nothing in their responses, are unable to see the present reality. For example, when I asked a client why she had confronted a colleague at work, she responded, “See you are just like everyone else in my life, you don’t understand me and you don’t like me, no one does and no one ever will.” This was the response of a blended firefighter, who, by my interpretation, lived in her brainstem or limbic system, where it embodied implicit memory. Clinicians and clients alike can learn to notice the ways in which these protectors and some exiles seem to come from the fight-flight-or-freeze brainstem region, or the highly emotional amygdala in the limbic system. Over time, I suspect we will learn where all parts are located in the brain.

Softening these implicit memory parts is particularly helpful for the goal of evoking Self-energy, the optimal environment for neural integration. I believe research will demonstrate that the method of IFS, which allows traumatized protectors and exiled parts to unburden toxic beliefs and feelings in the presence of the Self, is a way to deconstruct old pathways in the brain and is an effective strategy for converting implicit into explicit memory, paving the way for the formation of new neural networks. Once an unburdening has occurred, the brainstem, limbic system and cortical structures can work together in a newly integrated way.

NEUROBIOLOGY AND MEDICATIONS

The neurobiological effects of trauma on the body are widespread, affecting anatomical structures in the brain and neurotransmitter concentrations along with alterations in the endocrine and immune systems. High levels of stress have been shown to decrease hippocampal volume (Sherin & Nemeroff, 2011), an

area of the brain that is essential for explicit memory, and to increase firing in the amygdala, where implicit memory is encoded. It also reduces the volume of the prefrontal cortex, specifically the anterior cingulate (Sherin & Nemeroff, 2011), which normally calms down amygdala firing. Several neurotransmitters are altered by high stress conditions. There is a significant increase in norepinephrine levels and activity, thought to be a key factor in the development of posttraumatic stress disorder (PTSD), along with increases in dopamine and glutamate (Sherin & Nemeroff, 2011). Serotonin and GABA (gamma-aminobutyric acid) levels are decreased (Sherin & Nemeroff, 2011), both of which have calming effects on the brain. Malfunction in cortisol levels (high in acute trauma and low in chronic trauma) is common, as is a dysregulated hypothalamic-pituitary-adrenal axis (Sherin & Nemeroff, 2011). We also are aware of thyroid and endocrine abnormalities. Alterations in endogenous opioids may be associated with numbing, analgesia and dissociation (Sherin & Nemeroff, 2011). Thus, the body's response to stress is widespread and, under these conditions, it makes sense that the therapeutic process of IFS becomes difficult and neural integration and new pathway formations are hard to achieve.

To understand how medications play a role in restoring balance in the traumatized brain, we must take a brief look at individual nerve cells. Nerves in the brain mainly communicate by electrical impulse traveling down the cell to the synapse (the end) where vesicles contain neurotransmitters. These are released into the synaptic cleft and the neurotransmitters attach to a postsynaptic receptor, stimulating the electrical impulse by means of voltage dependent sodium or calcium ion channels along the next nerve cell (Schatzberg, Cole & DeBattista, 2010). Medications work by increasing or decreasing the concentration of neurotransmitters in the synapse, or stabilizing the electrical signal by affecting the voltage-dependent channels, promoting the proper communication between nerve cells. New technology allows us to measure neurotransmitter levels, which gives us a clearer picture of which levels need to be corrected (NeuroScience, 2009). Two types of neurotransmitters that medications affect are relevant to trauma: excitatory (glutamate and norepinephrine) and inhibitory (serotonin and GABA) (NeuroScience, 2009). In the excitatory category, glutamate is the primary neurotransmitter involved with learning and memory (NeuroScience, 2009). Too much glutamate is toxic for the hippocampus and can cause dissociation (Sherin & Nemeroff, 2011). Norepinephrine, which decreases in relation to depression, is primarily involved in arousal, alertness and emotional memory and in excess can contribute to the development of PTSD (Krystal & Neumeister, 2009). And, finally, dopamine (for the most part excitatory but can also be inhibitory) is involved in the reward (pleasure) circuitry connected to addiction as well as explicit memory and motor control (LeDoux, 2002). In the inhibitory category, GABA prevents overstimulation throughout the brain while the other inhibitory neurotransmitter, serotonin, modulates pain and libido. Serotonin also produces a sense of calm and contentment as it regulates norepinephrine and dopamine and can help reverse the atrophy in the hippocampus and pre-frontal cortex (Krystal & Neumeister, 2009). In acute trauma where there is an excess of excitatory neurotransmitters, psychopharmacology can help to calm the brain by decreasing excitatory neurotransmitters and increasing inhibitory neurotransmitters. PTSD can also be associated with periods of numbness and disconnection. Here we would be interested in increasing excitatory neurotransmitters to reverse the heightened inhibition of the brain. Research demonstrates that medications can stimulate neurogenesis (Schatzberg, Cole & DeBattista, 2010), and I believe that restoring the normal concentrations for neurotransmitters helps the brain reach the optimal state for neurogenesis and neural

integration, which enables a client to unblend more easily from protectors and exiles and to access Self-energy.

Now let us match medications with neurotransmitters. Some antidepressants increase the inhibitory neurotransmitter serotonin, others increase the excitatory neurotransmitters norepinephrine and dopamine and some do both. Often the goal with clients who have experienced trauma is to calm the brain and antidepressants that increase the inhibitory neurotransmitter serotonin are generally more helpful for this. Mood stabilizers, which decrease the excitatory neurotransmitter glutamate and increase the inhibitory neurotransmitter GABA, can also be useful for trauma. And they serve to calm the nervous system by affecting voltage channels and stabilizing nerve membranes. Anti-anxiety medications also increase GABA and can be helpful on an as-needed basis. Atypical neuroleptics, which decrease dopamine and affect serotonin (see [Table 7.1](#)), tend to be useful with trauma because they bind to the NMDA (N-methyl-D-aspartate) receptor, believed to be central to memory processing and inhibiting dissociation (Alderman, McCarthy & Marwood, 2009).

Table 7.1 Common Medications and the Neurotransmitters They Affect

- Serotonin & GABA (Gamma-aminobutyric acid)= Inhibitory Neurotransmitters
- Norepinephrine, Dopamine & Glutamate= Excitatory Neurotransmitters

ANTIDEPRESSANTS

- SSRI (Selective Serotonin Reuptake Inhibitor)
 - Prozac, Zoloft, Paxil, Celexa, Lexapro, Luvox (↑ Serotonin)
- Dual Agents
 - Effexor, Cymbalta, Remeron (↑ Serotonin, ↑ Norepinephrine)
 - Welbutrin (↑ Norepinephrine, ↑ Dopamine)

MOOD STABILIZERS

- Tegretol, Neurontin, Lyrica (Stabilize nerve membranes)
- Lithium (Stabilize nerve membranes, Balance neurotransmitters)
- Lamictal (Stabilize nerve membranes, ↓ Glutamate, ↑ Serotonin)
- Depakote, Topamax, Gabitril (Stabilize nerve membranes, ↑ GABA)

ATYPICAL NEUROLEPTICS

- Risperdal, Zyprexa, Seroquel, Geodon
 - (↓ Serotonin 2a (side effects), ↑ Serotonin 1a (calming), ↓ Dopamine)
- Abilify
 - (↓ Serotonin 2a (side effects), ↑ Serotonin 1a (calming), ↑ or ↓ Dopamine)

BENZODIAZEPINES

- Ativan, Klonipen, Xanax, Valium, Serax, Librium (↑ GABA-A)

ANTIADRENERGICS

- Clonidine, Inderal, Minipress, Tenex, Intuniv (↓ Norepinephrine)

STIMULANTS

- Ritalin, Concerta, Adderall, Vyvance, Metadate, Focalin
 - (↑ Norepinephrine, ↑ Dopamine)
- Strattera (↑ Norepinephrine)

NONBENZODIAZEPINE SLEEP AIDS

- Ambien, Sonata, Lunesta (↑ GABA-A alpha 1 receptor)

Source: Physicians' desk reference, [2012](#); Schatzberg, Cole & DeBattista, 2010.

Several studies have shown the effects of high levels of norepinephrine on the pathogenesis of PTSD (Sherin & Nemeroff, 2011). On the one hand, anti-adrenergic drugs, which decrease the excitatory neurotransmitter norepinephrine, are useful for treating acute trauma as well as intrusive memories and flashbacks (Alderman, McCarthy & Marwood, 2009). On the other hand, stimulants, which increase low levels of norepinephrine and/or dopa-mine, are useful to “activate” the prefrontal cortex in cases of numbing and disconnection that develop over time. Stimulants are not, however, advisable for acute trauma. Medication for trauma clients can be chosen according to its effect on inhibitory and excitatory neurotransmitters. Optimizing neural integration by balancing neurotransmitters opens the door, in IFS language, to unblending extreme parts and bringing Self-energy to the fore.

WORKING WITH MEDICATIONS AND PARTS IN TREATMENT

Richard Schwartz (personal communication, August 2011) said, “Parts can push the biological button.” I agree and I also have seen the reverse, that biology can affect parts. If a client is not on medication and feels stuck with parts who cannot unblend, then consider a medication consult because it is a good unblending agent. If, on the other hand, a client has tried several medications and none have worked or she complains of multiple side effects, direct communication with parts about medication decisions is a good policy. The clinician can query the client's internal system of parts about any responses to medication, or ask a specific part how it feels about taking medication in general or wonder how the part may be reacting to a specific medication. I believe that parts need to be included in the decision-making process for a medication to work properly. Ideally, the prescribing clinician should do this work but

many prescribers are not familiar with parts work. In this case, the therapist can help the client understand the feelings of her parts about medications and how to *speak for them* when seeking medication support. Here is an outline of my method for working on medication with parts.

First, I open a dialogue with parts around medication decisions and help the client to *identify* a list of symptoms for treatment. Remember medications *treat symptoms not feelings* and medications work by correcting the neurotransmitters, imbalances (which cause symptoms) thus bringing a client's system back into equilibrium. It is important for the clinician and the client to be clear which symptoms they want to treat and to be able to track progress over time. Low energy, reactivity, panic and numbness, for example, are symptoms while sadness, anger, loneliness and despair are feelings. Parts can help differentiate a psychological (part response) from a biological condition. Biological conditions that can be associated with trauma are PTSD, dissociation, depression, substance abuse, eating disorders, attention deficit disorder and obsessive-compulsive disorder, to name a few. When symptoms related to these diagnoses occur it is important to determine if they are purely biological in origin or if they come from, for example, a depressed part who is protecting an exile by withdrawing, isolating and disconnecting; or a frightened, overwhelmed part who pushes the biological button and causes a panic attack. When asked, parts are often helpful in determining the relationship between a part and biology. In my experience, they are often clear about whether a part is causing a symptom or the system of parts is experiencing the symptom as something happening to them through a biological process— when in doubt, ask the part.

Once you have determined the origin of the symptoms, it is important to *validate* the experiences of the client's parts as well as addressing any concerns they have about taking medications. When parts feel validated they are much more likely to be cooperative with a medication trial. I will not prescribe medications unless all parts are in agreement. However, some obstacles to reaching consensus are common. Hypervigilant parts can interfere. As well, differing responses among parts to medication can be a problem. When this happens, I make sure the client has heard the feelings of all parts and that the parts feel the client understands. For example, there may be a part who feels exhausted and desperately wants relief along with a part who is afraid to take anything into the body and a third part who needs more information to feel comfortable. It is important for all parts to feel the client really hears them and will proceed only after taking their view into account. And polarizations, which can motivate parts to affect or block medication, often surface when we explore parts' fears. When I notice a polarization between parts, I explore the feelings, opinions and agendas of both sides. In my experience, this often leads to an exile, and the polarization around medication decisions must be addressed before the exile can be unburdened.

Finally, I *educate* parts about what they can expect from a medication. I invite all parts to communicate their experience and concerns, and I promise that the client and I will maintain an open dialogue and *collaborate* with them throughout the process. In addition, I am mindful of my own parts whenever they interfere with me prescribing from a place of Self-energy. I find that my helper and helpless parts come up frequently. My frame of *I educate, you decide* helps my parts let me stay in Self-energy. Here are several case examples illustrating various ways in which biology can affect parts and vice versa.

BIOLOGY AFFECTING PARTS

Nina, whom I saw for both therapy and medication management, was going through the break-up of a long-term relationship. Feeling consistently overwhelmed and blended with a little girl part, she did not believe she had the resources to endure her pain.

“Let’s check in with your parts,” I suggested.

“The little girl feels like there is a thick layer of Plexiglas between us and it’s keeping her from feeling connected to me,” Nina reported. “She’s frozen and sad.”

“Is the Plexiglas a part?” I asked.

“No. I’m getting ‘no’ on that,” she said.

“Are all of your parts feeling the same thing?” I asked.

“Yes,” she said. “And they are missing the connection with me.”

“Ask them if they all want some help with this,” I said.

“Yes, they all want help.”

“Ask if they are willing to take a medication to help make the Plexiglas go away,” I said.

“Yes they all are willing,” Nina reported.

“I think an SSRI will give your body the best help with the Plexiglas feeling,” I said.

After educating her about what to expect from the medication, including potential side effects, I gave her clear instructions on how to increase the dose slowly. Then I asked her to check in with her parts again to see if they had any questions and if they still wanted to proceed. All her parts said yes. “If they have any concerns at any point in the process, ask them please to let you know. And promise them that we will address their concerns,” I said.

After agreeing to a brief weekly check in to see if she noticed any changes or heard any concerns from her parts, Nina left with a prescription. After the third week of using the medication she reported that she could feel it starting to work. She felt lighter and was able to hear from her parts more easily. I asked her to check with her parts to see how they were feeling. Tears came to her eyes as she heard them chiming in about how pleased they were to feel better and how grateful they were that someone actually cared enough to ask them how they were feeling, which they were not used to. They described being able to see and feel her presence more clearly.

In this example, a medication was given with all parts participating in the decision, the medication helped and therapy could proceed. Generally speaking, medications work better when all parts instead of only one part are affected by the symptoms and when all parts agree to take a medication. In this case, a biological condition was affecting a client’s parts and interfering with the process of therapy.

PARTS AFFECTING BIOLOGY

Sue had been in therapy for several years and had done remarkably well over all. She had a significant trauma history including sadistic sexual abuse by her cousin. Over the spring she had been feeling much

better. She had been coming twice a week for approximately 6 years and was thinking of cutting down on the frequency of her visits. This seemed reasonable to me. We agreed to finish a piece of work with her 12-year-old girl and then she would change to weekly visits by summer.

As work with the 12-year-old progressed, Sue's behavior changed. She cancelled appointments for the first time and stopped paying her bills consistently, developing a running balance. She seemed increasingly depressed although she remained on antidepressants. She gained 10 pounds and lost motivation to connect with friends. It was even difficult to discuss these changes in her behavior due to her inconsistent attendance. I knew something was up but did not know what it was. This pattern of atypical behavior went on for weeks before I was able to bring it up.

"Sue," I said, "are you sure there isn't something going on with any of your parts? We both know that missing appointments and not paying your bills is not typical for you. Really, take time to ask inside if any parts have anything to tell you about this?"

Sue sat in silence, checking inside, and then began to cry. "I am so afraid you are going to hate me," she said.

"What are you talking about?" I asked.

"It is my twelve-year-old. She has been so afraid to deal with this last piece of work. She doesn't want to come to therapy." Sue continued to weep. "She feels so ashamed. She likes you so much but she's afraid to go on. This is the only way she knows to make things slow down."

"I understand completely," I said. "That makes perfect sense. Tell her that we can slow things down and I'm sorry I didn't hear her sooner." I added, "Ask if she has anything to do with you feeling depressed and withdrawn?"

"Yes definitely!" Sue said. "She's been feeling so bad and not knowing what to do to fix it."

"Ask her if she wants help with that," I said.

"She wants help and she wants to make sure you're not mad at her," Sue reported.

"Tell her that I'm not mad. I'm grateful that she is now able to let us know what she feels. Ask her if the medication is still working and, if so, does she want to increase it?"

"She says the medicine is not working and she would be willing to try another one," Sue reported.

"Check and see if all parts are OK with a change in medication," I said.

"Yes," Sue said. "They all want to feel better."

In this example, a 12-year-old part who was terrified of paying attention to her experience tried to slow the therapy down in the only ways she knew, including pushing the biological button to increase Sue's depression even though she was already on an antidepressant. Once Sue connected with this part in a Self-led fashion the part responded, Sue listened to the part and changed the medication. The depression lifted within a couple weeks and the atypical behavior around appointments and payment resolved immediately.

This example illustrates an exile pushing the biological button because it was feeling totally overwhelmed. The part slowed the therapy down by missing appointments, not paying bills and increasing Sue's depression. She was afraid this behavior had made me angry but after validation and reassurance, she was able to request a medication change that helped her deal with overwhelming feelings and the therapy could proceed.

VALIDATING AND ADDRESSING CONCERNS

As mentioned, I will not prescribe medication unless all parts agree. In this way, though the client may be unaware of them, I am able to address concerns that parts may be carrying. Protective parts often feel suspicious about the client's motive for taking medication and fear being eliminated. Once these feelings have been expressed and validated, the client's Self can explain the actual goals of taking medication, which includes helping—not eliminating—hard-working protectors.

Tom, a client who was in IFS therapy with another therapist, came to me to discuss medication. He had been on a selective serotonin re-uptake inhibitor (SSRI) for social anxiety for several months with no significant improvement.

I said, "Let's check with your parts about the Paxil you are taking to see if there are any parts who may feel resistant to taking it."

"Hmmm," Tom said with a puzzled look. "I asked my little boy about the anxiety and medication and he doesn't even know what anxiety is."

"Check to see if any other parts do know about the anxiety and medication," I said.

"There seems to be a part that is older, like a teenager that wants to keep me anxious!" Tom said in surprise.

"Are you open to hearing more from this part?" I said.

"It says the anxiety is helping. If I am feeling better I will start going out with friends, meeting new people, and they will see how inadequate I really am."

"Does this make sense to you?" I said.

"Well, yes it does but I had no idea a part was feeling this way," he said.

"Do you understand it now?" I asked.

"Yes," he said.

"Ask the part if it has had anything to do with the medication not working," I said.

"Yes it has been blocking the medication," he said.

"Does the part feel like you understand where it's coming from and why it's doing this?" I said.

"I think so," he said.

"Ask if we can help find other ways of protecting you from getting hurt so it doesn't have to work so hard," I said.

"It likes the idea of not working so hard," he said.

"Good. Let it know that the medication can also help it not work so hard," I said.

"It's open to that," he said.

"Tom," I said, "I need you to be committed to listening to this part and its fears as we move forward with medication. Are you willing to do that?"

"Yes I am," he said. "The part really likes that idea."

So Tom got permission from the part to increase the Paxil by 12.5 milli-grams, and he assured the part that he was open to hearing any of its concerns. At the follow-up appointment he was significantly less anxious.

In this example, a manager was blocking the positive effects of medication to protect an exile who was burdened from being hurt and humiliated in social situations. Validating the manager and addressing its fear helped it feel willing to relax and cooperate with Self in allowing the medication to decrease his anxiety.

POLARIZATIONS

Parts often become polarized around medications. One part wants relief while other parts fear, dislike or disapprove of medication. A history of failed medication trials is a red flag for polarized parts, who can override medication responses when they do not feel heard. Carol came in to consult about medications for trauma and dissociation. She was in an IFS therapy but no one had ever asked her parts how they felt about the various medications she had been taking over many years. When she checked with her parts on their feelings about medication, she was surprised to hear, “Can’t get better; can’t get better!”

She asked why and the part replied, “If we get better we will get beat.”

“Beat by who?” I asked.

“My mother,” Carol sighed. “Mom has always been threatened by us doing well.”

“Does this make sense to you? If so let this part know,” I said. After she had done so, I went on, “I suspect that this part has something to do with the medications not working well for you. Let’s also check to see if any other parts have feelings about medication.”

Now I am hearing, “*We always do what the people in charge say. I am a good little girl. Wow!*” Carol said. “Now I understand why I’ve always taken anything anyone has ever wanted to give me even though nothing ever helps!”

“Let her know that you get how important it is for her to be a good girl,” I said.

After Carol understood the root of her complicated medication history she was able to begin working with her parts more directly around medication decisions. When she got consensus to try a low dose of Abilify for dissociation, we proceeded.

“Make sure that both the scared girl and the good girl understand that the goal of this medication is to help them so they don’t have to work so hard against each other. Let them know you’re not trying to get rid of either one of them,” I said.

“I think they get it now,” she said.

This example illustrates how polarizations can complicate medication responses and choices. Not listening to parts had resulted in a long history of failed medication trials. A polarization was revealed. One part was afraid of medications working while the other part would take any medication given. This dilemma was resolved by the client’s Self validating the parts and offering them the option of not having to work so hard.

MANAGER VERSUS FIREFIGHTER RESPONSES

Specific categories of parts tend to respond differently to medication. Managers are proactive protectors whose mission is to avoid emotional pain while firefighters are reactive protectors whose mission is to douse emotional pain. Firefighters tend to be absolute and severe (an example of living in implicit memory). They often override medication responses out of fear of being made to disappear or lose their job. They develop trust slowly and may only be willing to take medication as needed (known in the medical world as PRNs). Atypical neuroleptics and benzodiazepines, which decrease severe reactivity and anxiety, can be taken as needed. These medications are good choices when negotiating with a firefighter, however with addictions atypical neuroleptics are preferred. At times it may be necessary to use direct access to learn about the feelings and beliefs of a particular part, but enough Self-energy must be available to facilitate working with and hearing from all the client's parts so they can reach consensus. As the next example illustrates, in cases of severe trauma it may take time for protective parts to unblend so that Self-energy is available.

Phoebe, who had dissociative identity disorder (DID), originally came for a medication consult. She had been to several psychiatrists over the years and in our first session she stated, "Medications do not work for me, I have tried everything and I can't tolerate anything."

"Do you know how your parts feel about taking medications?" I asked.

She said, "I have no idea. No one has ever asked them before."

I said, "Is it OK to ask them now?"

Phoebe went inside and pretty quickly heard from a part who said, "You know you can't trust doctors. They are all the same."

The part went on to remind her that the therapist who had sexually abused her in graduate school kept pushing her to go to a psychiatrist and take more pills. She believed he wanted her to be in an altered state so that she would be more compliant with the abuse.

"Boy, that makes a lot of sense doesn't it?" I said.

"Yes I guess it does," she said.

"Let this part know that you get its fear about taking medications," I said. "Also let it know that you will not take any medication until it trusts us and we have its total permission."

Over the next 4 months Phoebe and I met weekly and she decided to transfer her therapy to me. Her skeptical part also began to trust me and eventually it was willing to try a very small dose of Seroquel 6 milligrams (compounded by a pharmacy). Hypervigilant parts are often associated with physiologic hypersensitivity and so tend to respond well to "microdoses" of medications. Standard doses of medications are often associated with intolerable side effects that can frustrate a prescriber who is unaware of this and leave a client feeling misunderstood. Some parts are resistant to traditional medications and should be educated about alternative treatments such as herbal remedies, natural

products or acupuncture. Products that naturally support neurotransmitter production are also available. Two such examples are: 5-Hydroxytryptophan (5-HTP), a serotonin precursor, and phosphatidylserine, which can restore cortisol receptor sensitivity and help HPA axis communication.

The part was willing to try the medication but wanted permission for Phoebe to e-mail me with any questions. It also wanted to try the medication on the weekend when not much was going on. Phoebe said, “That is all fine with me.”

“Check in to see if all parts are OK with this plan,” I said.

“I don’t hear any complaints,” she reported.

Managers, in contrast to firefighters, are often willing to admit exhaustion and seek relief. They will usually take medication and they like the sense of distance and perspective that can be created by selective serotonin re-uptake inhibitors (SSRIs). Therefore, at least for these parts, SSRIs seem like good unblending agents. In addition, exploring medication options serves the helpful function of revealing polarizations between managers, firefighters and/or exiles. In this example, both client and prescriber worked with a vigilant protector to alleviate its fears, address its concerns and develop a trusting relationship.

KEEPING COMMUNICATION OPEN

Parts appreciate knowing what to expect from a medication and can be helpful in assessing dosage. They can also help by tracking and classifying physical sensations that could be caused by the medication or could, instead, be the communication of a part. Here is an example of the importance of keeping communication open and paying attention to sensations that arise in relation to medication.

Sean came in for a psychopharmacology appointment saying he felt good and wanted to find out if he could maintain this improved mood on a lower dose of medication.

“Ask your parts how they feel about tapering the Lamictal,” I said.

“I am noticing some tension and tightness in my shoulders right now,” he observed.

“Would it be OK to focus on that sensation and ask what’s up?” I said.

“Well, there is a part that’s worried about tapering,” he said. “It says there is a lot going on right now. My wife is particularly stressed and I am getting a new boss at work. This part thinks it’s not the best time to make changes.”

“How do you feel toward the part?” I asked.

“It makes sense,” he said. “I’m glad the part spoke up but I still want to taper.”

“Let this part know you really get what it is telling you and ask what would help to make it feel comfortable with you tapering,” I said.

He closed his eyes for a moment and then said, “If I promise to listen and pay attention then it will feel OK about trying. It says I need to listen to the tension in my shoulders.” So they agreed to stay in close communication.

“Check to see if any other parts have concerns,” I said.

“No I don’t think so. I never knew my muscle tension was a part,” he said. “That surprises me. I feel good that my parts can help me make this decision. I am more confident moving forward with the taper.”

This example illustrates the importance of listening to and working with a manager who communicates through the body. Here a part expressed fear about tapering medications due to stressors in the client’s life. After listening to the body, his parts agreed to taper more slowly if the client would agree to listen to them along the way.

STAYING IN SELF, ADDRESSING THERAPIST'S PARTS

Just like any IFS therapist, the prescribing clinician must work with his or her parts. Both my helper parts and my helpless parts can be activated when I prescribe medications. When I experience a dilemma about whether to witness a client’s pain or prescribe a medication to reduce or remove a symptom, my helper parts tend to get activated. A therapist who is considering whether to send someone for a psychopharmacology consultation could have a similar dilemma. My helpless parts tend to get activated when parts of the client feel frustrated because medications are not working.

Paul had been feeling disconnected lately and believed he was depressed. He had a history of bipolar disorder, which was well controlled with a mood stabilizer. He came into a session saying he wanted to try an antidepressant as well.

“Check in with your parts to see how they feel about this,” I said.

“No response—all quiet inside,” he said quickly.

He really wanted an SSRI so I agreed and he started on a low dose, but over the next several weeks he became numb and felt worse. At this point I realized that a part had been activated in me. This part had agreed too easily with Paul’s wish to take an SSRI. When I turned my attention inside, I heard a young part say, “If I give him what he wants he will like me,” and I realized I was blended with my helper. I knew if this part had stepped back I would have been more curious about Paul’s statement, “No response—all quiet inside,” and I would have asked about his urgency to try a new medication.

I asked my helper to relax now and let me be curious. As it agreed, I said to Paul, “Check in with your parts again and see if we can learn anything about what is going on here.”

On going inside, much to his surprise, he discovered a new part. He said, “I think it’s a numbing part.”

“See if this part can tell you more about its job,” I suggested.

“It says it is protecting me from having any feelings. It has been around for a while and does not want me to have any feelings about my sister’s new diagnosis of leukemia. It says to me that I will never be able to handle it if anything happens to her.”

“Does this make sense to you,” I said.

“Yes totally,” he said.

“Let this part know you understand and ask it if it would like some help or relief with a medication,” I said.

“It says no, it feels glad that I finally know that it exists.”

“Check in with other parts now to see if it’s OK to taper off the medication,” I said.

“Yes totally.”

We planned a taper schedule and he began to work more directly with the part. Taking this new tack, his numbing protector softened and he was able to get to know an exile who had taken care of his younger sister when they were kids. This exile felt that his sister only had him, and conversely, he only had his sister so she was essential to his survival.

Here we see that a symptom that initially looks biological can turn out to be a protector. In this case a numbing protector was trying to keep Paul from feeling deep pain about the neglect and losses that he and his sister suffered in childhood as well as the coming loss of his vital connection to her. My helper part had gone along with his protector and had gotten in the way of my curiosity about his silence and the urgency he felt to take a medication. I discovered all of this only after he started the medication (which was biologically unnecessary) and noticed that he felt overmedicated. This example illustrates how a part in the prescriber can interfere, allowing a medication to be prescribed which was unnecessary and proved unhelpful.

MY PART FEELS PUSHED

John and Michael come in for their usual appointment, however this particular week John seemed more agitated than usual. “I can’t take it anymore,” he said. “Michael is so depressed and unable to get anything done at home. I feel like I am doing everything! I work all day and then have to come home at night, cook dinner and clean the house. He does nothing all day long but sit on the couch playing games on the computer. We have been coming here for a while now,” John looked at me fiercely. “You are a psychiatrist. Why don’t you give him a pill or something to stop the drinking?”

I looked at Michael. “How do you feel about what John just said?” I asked.

“I am tired of being criticized by him,” Michael said. “I can’t do anything right. I think I would drink a lot less if he would back off. I’m not that bad.”

“John,” I said, “I think Michael might be able to take a look at his drinking if he felt less criticized.”

“I knew it!” he said. “His drinking is my fault. Once again the responsibility lies on my shoulders. It always turns out that way, my whole life. I want out!”

Aware at this point that a part of mine had incensed a part of John, I took a moment to check inside. I quickly found a part who felt pushed by John into trying to fix Michael. And this part did not like being told what to do.

Taking a deep breath, I said, “John, I hear how upset and frustrated you are about Michael’s drinking. And I hear that a part of you would like me to prescribe a medication to help him stop and function better at home. That makes a lot of sense to me. And Michael you seem upset and shut down by what you describe as John being critical and attacking. And you don’t really feel that your drinking is a

problem. I cannot tell if a part of you is drinking in reaction to John's criticism or there is a biological addiction we are working with. But before we discuss that, let me educate you both about some of the options available for alcohol treatment, including medications. Revia is a medication that reduces cravings, Antabuse treats binge drinking and Clonidine helps with withdrawal symptoms. I am here to answer your questions every step of the way." They looked at me and then at each other, their faces softening, and I proceeded to explain their options.

In this example, once I was able to connect with my reactive part who felt pushed into doing something, it was able to relax and I could proceed to the job of assessing and possibly prescribing with Self-energy. I generally find that the stance *I educate, you decide* elicits Self-energy in the client and allows my parts to step back.

CONCLUSIONS

In summary, I have developed five strategies to integrate psychopharmacology for trauma with the basic principles of IFS therapy. First, I identify a list of symptoms; second I differentiate a part response from a biological condition; third, I validate the experience of parts and address their concerns, with a policy of only prescribing when there is internal consensus about trying medication; fourth, I educate parts about what to expect from medications and collaborate with them throughout the prescribing process and, fifth, as I guide and educate the client (who makes the decisions), I am mindful of my own parts.

Neuroscience shows that the brain is capable of changing (neuroplasticity) and developing new nerve cells (neurogenesis), which ultimately result in the formation of new pathways and behavioral change. For the formation of new networks, certain parts of the brain need to be working together (neural integration). In trauma, the physiological effects of stress cause activation in the brainstem, increased amygdala firing and hippocampal atrophy in the limbic system and hypothalamic-pituitary-adrenal dysfunction. Stress also affects the cortex, resulting in decreased volume in the prefrontal cortex and atrophy of the anterior cingulate. One of the ways to correct abnormalities in brain function is to restore neurotransmitter concentrations with medications. I believe these abnormalities cause symptoms, which can block treatment, and that correcting them allows IFS therapy to progress more efficiently.

To work at the interface of biology and psychology, I open a dialogue with parts that is geared to differentiate the physical from the emotional and cognitive. Medications treat symptoms not feelings and when asked, parts will describe a biological process that "happens to them," or will describe themselves feeling overwhelmed and fearful and "pushing the biological button." Validating the concerns of parts about medication helps them to feel understood and be willing to cooperate with a medication regimen. When not included in the decision-making process, parts can block the beneficial effects of medication. It is also common for polarizations to develop, with different parts having different feelings about a particular medication. When the client can hear both sides with Self-energy, a consensus will often emerge. On the whole, medications help to support and relieve hard-working protectors.

Typically, different medications are helpful for different symptoms following trauma. Knowing which neurotransmitters are affected by which medication can help prescribers to determine which

symptoms can be treated. Generally SSRIs, which work by increasing the calming neurotransmitter serotonin, are helpful for symptoms related to PTSD. Mood stabilizers help by stabilizing nerve membranes and increasing the calming neurotransmitter GABA and decreasing the excitatory neurotransmitter glutamate. The atypical neuroleptics tend to be helpful for dissociation because they affect serotonin, block dopamine and affect the NMDA receptor. Antiadrenergic medications, which decrease the excitatory neurotransmitter norepinephrine, are helpful for intrusions, flashbacks and acute trauma. Benzodiazepines work by increasing the calming neurotransmitter GABA and should be only used as needed due to their addictive potential. They are, however, helpful for treating acute anxiety. Last, stimulants can be used for trauma when an increase of dopamine and norepinephrine is needed to activate the prefrontal cortex while treating numbness or desiring an increase in focus and attention.

Just as different medications treat different symptoms, different parts tend to respond to different medications. Firefighters with their reactive all-or-nothing responses come from implicit memory and, most likely, originate from more primitive brain structures such as the brainstem and limbic regions. Firefighters typically take a long time to develop trust with Self and will often only take medications on an as-needed basis. They tend to respond to atypical neuroleptics and benzodiazepines. Managers, on the other hand, are often more willing to take medications because they want relief. They tend to benefit from SSRIs and mood stabilizers. Hypervigilant parts tend to respond well to microdoses of medications. They also like natural remedies that are less potent and tend to reject sleep medications, which thwart their goal. Educating parts and keeping communication open throughout the medication process is vital for developing trust between Self and parts.

The connection between neuroscience, psychopharmacology and IFS is an exciting frontier. In my opinion, correcting chemical imbalances in the brain by eliciting the help of parts in the process of prescribing medications paves the way for protectors to step back and Self-energy to emerge so that exiles can unburden (old pathways are deconstructed) and new neural networks can form in the brain.

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